

Olympiad Test Paper — Science (Class 7)

Chapter 16: Water: A Precious Resource

Subject	Science (NCERT Class 7)
Chapter	16 — Water: A Precious Resource
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. All questions stay within the Class 7 chapter — the three forms of water, the water cycle, where Earth's water is stored, the water table and aquifers, why the water table rises or falls, water scarcity, and water management. No calculator is needed. Most questions need two to four reasoning steps; watch for traps — Earth has lots of water, but very little of it is usable fresh water. Do not write on this paper — mark answers on your answer sheet.

1. Water exists on Earth in three forms. Which set lists all three correctly?

(A) ice, rain, snow

(B) ice, liquid water, water vapour

(C) liquid water, steam, clouds

(D) ice, dew, fog

2. When wet clothes dry in the Sun, the liquid water turns directly into:

(A) ice

(B) water vapour

(C) clouds

(D) rain

3. Which process in the water cycle is **mainly powered by the Sun**?

(A) condensation of vapour into cloud droplets (B) precipitation as rain

(C) evaporation of water from oceans and land (D) infiltration of rain into the soil

4. Dew forms on grass on a cold morning. This is an example of:

(A) evaporation

(B) precipitation

(C) infiltration

(D) condensation

5. In the water cycle, clouds are formed when:

(A) rain falls back into the ocean

(B) liquid water freezes into ice on the ground

(C) rivers carry water to the sea

(D) water vapour high in the cool air condenses into tiny droplets

6. Plants release water vapour into the air from their leaves. This step of the water cycle is called:

- (A) evaporation
- (B) condensation
- (C) transpiration
- (D) precipitation

7. Rain, snow and hail are all grouped under one stage of the water cycle. That stage is:

- (A) evaporation
- (B) condensation
- (C) transpiration
- (D) precipitation

8. The total amount of water on Earth, over the years, stays roughly the same because water in the cycle is:

- (A) only changing its form and location, not its total amount
- (B) constantly being created by rain
- (C) constantly being destroyed by evaporation
- (D) slowly leaking into outer space

9. Put these water-cycle events in their natural order: precipitation, evaporation, condensation.

- (A) condensation → evaporation → precipitation
- (B) evaporation → precipitation → condensation
- (C) precipitation → evaporation → condensation
- (D) evaporation → condensation → precipitation

10. About what fraction of all the water on Earth is in the seas and oceans?

- (A) about 97%
- (B) about 50%
- (C) about 2%
- (D) about 0.5%

11. Sea water cannot be used directly for drinking or for watering crops mainly because it is:

- (A) too cold
- (B) frozen
- (C) too deep
- (D) salty

12. A student says, "Earth has plenty of water, so we can never run short of water to drink." The best correction is:

- (A) the statement is fully correct
- (B) Earth actually has very little water of any kind
- (C) most of Earth's water is salty or frozen, so usable fresh water is very limited
- (D) all sea water is fresh and drinkable

13. Most of Earth's **fresh** water is not easily available to us because it is:

- (A) all in the air as vapour
- (B) frozen in ice-caps and glaciers or stored deep underground
- (C) entirely inside plants
- (D) locked inside ocean salt

14. Which statement about Earth's water is correct?

- (A) most of Earth's water is fresh and ready to drink
- (B) there is no salt water on Earth

- (C) only a tiny fraction of Earth's water is fresh water we can readily use
- (D) glaciers contain no fresh water

15. India has about 17% of the world's people but only about 4% of the world's fresh water. This tells us that India:

- (A) has more than enough water for everyone
- (B) has no fresh water at all
- (C) is under water stress, with limited fresh water for a large population
- (D) should export its fresh water

16. The water that fills the spaces between soil particles and rocks below the ground is called:

- (A) ground water
- (B) surface water
- (C) sea water
- (D) rain water

17. The **water table** is best described as the:

- (A) flat top of the ocean
- (B) upper level of the water stored underground
- (C) deepest river in a region
- (D) layer of clouds before rain

18. After a long spell of good rain over a region with open soil, the water table is most likely to:

- (A) rise
- (B) fall
- (C) disappear
- (D) turn salty

19. The seeping of rain water down through the soil to reach the ground water is called:

- (A) evaporation
- (B) condensation
- (C) transpiration
- (D) infiltration

20. An **aquifer** is best described as:

- (A) a large surface lake
- (B) a machine that pumps water
- (C) a cloud full of rain
- (D) a layer of porous rock or gravel underground that stores water

21. Ground water is replenished ("recharged") mainly when:

- (A) rain water infiltrates through the soil into the aquifer
- (B) factories release waste water into rivers
- (C) people pump water out for irrigation
- (D) the Sun evaporates water from the sea

22. In a village, the amount of ground water pumped out each year is greater than the amount that seeps back in. Over several years the water table will:

- (A) stay exactly the same
- (B) keep rising
- (C) keep falling
- (D) rise then fall back to the start

23. Which everyday observation is the clearest sign that the water table in an area has fallen over the years?

- (A) the village pond fills up after rain
- (B) clouds form in the sky before the monsoon
- (C) dew collects on the grass each morning
- (D) hand-pumps and wells that once gave water now run dry

24. Which of the following would tend to **lower** the water table?

- (A) pumping out large amounts of ground water for irrigation
- (B) building recharge pits in the colony
- (C) heavy, steady monsoon rain on open ground
- (D) planting many trees

25. A town paves over its open grounds and parks with concrete. The most direct effect on ground water is that:

- (A) less rain water can soak in, so recharge falls and the water table tends to drop
- (B) more rain water now infiltrates the soil
- (C) the water table rises faster
- (D) the rain becomes salty

26. Which of these is **not** a cause of the falling water table?

- (A) a rapidly growing population needing more water
- (B) covering large areas with roads and concrete
- (C) heavy rainfall soaking into open soil
- (D) intensive irrigation drawing a lot of ground water

27. Cutting down forests (deforestation) tends to lower the water table because, with fewer trees:

- (A) less rain water is held back and allowed to soak in, so recharge drops
- (B) the Sun stops shining
- (C) the rain becomes salty
- (D) rivers stop flowing entirely

28. Which combination of changes would push a region's water table **down** the fastest?

- (A) good rainfall, many recharge pits, and forests
- (B) rising population, more industry, and heavy paving with scanty rainfall
- (C) drip irrigation and rooftop rainwater harvesting
- (D) planting trees and mending leaks

29. Two neighbouring colonies get the same rainfall. Colony X has open soil and gardens; Colony Y is fully concreted. After the monsoon, the water table is likely to be higher under:

- (A) Colony Y, because concrete traps water
- (B) neither — concrete and soil behave the same
- (C) Colony Y, because gardens use up water
- (D) Colony X, because open soil lets rain infiltrate and recharge the ground water

30. Which single action would best help **raise** a falling water table in a city?

- (A) paving over all the parks
- (B) pumping more ground water for swimming pools
- (C) building rainwater-harvesting recharge pits so rain soaks into the ground
- (D) cutting down street trees

31. A farmer switches a field from flood irrigation to drip irrigation. The effect on ground water is that the field now:

- (A) wastes much more water, lowering the water table faster
- (B) makes the ground water salty
- (C) needs no water at all
- (D) uses much less water, easing the demand on ground water

32. Among the common irrigation methods, the one that **wastes the most water** is:

- (A) flood (surface) irrigation
- (B) sprinkler irrigation
- (C) drip irrigation
- (D) watering each plant by hand

33. Drip irrigation saves water mainly because it:

- (A) sprays water high into the air over the whole field
- (B) waters only at night
- (C) floods the entire basin
- (D) delivers water drop by drop directly to each plant's root zone

34. A gardener wants to water a row of tomato plants using the **least** water. The best method is:

- (A) flooding the whole bed
- (B) a sprinkler running all day
- (C) a drip system delivering water at each plant's roots
- (D) hosing the leaves from above

35. "Rainwater harvesting" means:

- (A) buying bottled water in the rainy season
- (B) pumping ground water faster during the rains
- (C) draining all rain water quickly into the sea
- (D) collecting and storing rain where it falls, or letting it soak in to recharge ground water

36. A household leads the rain falling on its rooftop through a pipe into an underground tank. This is an example of:

- (A) drip irrigation
- (B) rooftop rainwater harvesting
- (C) deforestation
- (D) flood irrigation

37. A recharge pit dug in a colony helps the water table because it:

- (A) lets collected rain water seep down into the aquifer
- (B) pumps water out of the ground
- (C) seals the soil with concrete
- (D) evaporates water into the air

38. A bawri (step well) and a tanka are traditional Indian structures used for:

- (A) generating electricity
- (B) cooking food
- (C) measuring rainfall
- (D) storing or harvesting rain and ground water

39. After a city makes rooftop rainwater harvesting common, its ground water table over a few years tends to:

- (A) fall sharply
- (B) become salty
- (C) rise, because more rain is directed into the ground instead of running off
- (D) stay frozen

40. World Water Day, observed to remind people about saving water, falls on:

- (A) 5 June
- (B) 1 January
- (C) 14 November
- (D) 22 March

41. During a severe water shortage, garden plants that do not get enough water will most likely:

- (A) wilt — their leaves and stems droop
- (B) grow taller and greener
- (C) turn into vapour
- (D) start flowering more

42. Water scarcity affects people most directly by:

- (A) giving everyone more drinking water
- (B) cooling the whole town
- (C) raising the water table
- (D) making it hard to get enough water for drinking, cooking, washing and crops

43. A dripping tap can waste around 30 litres of water in a single day. The best response is to:

- (A) ignore it, since one tap is too small to matter
- (B) open the tap fully so it stops dripping
- (C) mend the leak promptly to stop the waste
- (D) leave a bucket forever under every tap

44. Which everyday habit saves the **most** water during a bath?

- (A) a long shower
- (B) leaving the tap running while soaping
- (C) using a bucket and mug instead of a long shower
- (D) filling the bathtub to the top

45. Closing the tap while brushing your teeth helps because it:

- (A) makes the toothpaste work better
- (B) stops clean water from running away unused
- (C) raises the sea level
- (D) speeds up the water cycle

46. Which set of actions all help **conserve** fresh water?

- (A) rainwater harvesting, drip irrigation, mending leaks
- (B) paving open ground, flood irrigation, leaving taps running

- (C) cutting forests, draining ponds, long showers
- (D) over-pumping wells, concreting parks, ignoring leaks

47. A school wants to reduce its demand on ground water. Which plan is best?

- (A) drill more bore wells and pump harder
- (B) harvest rooftop rain into a recharge pit and fix all leaking taps
- (C) pave the playground to keep it clean (D) leave the garden sprinklers on overnight

48. Which statement correctly compares water amounts on Earth?

- (A) fresh usable water is far more than ocean water
- (B) glacier ice is the largest source of drinking water we use daily
- (C) all the water is fresh
- (D) ocean (salty) water is far more than the fresh water we can readily use

49. Ice, liquid water and water vapour are:

- (A) three different substances
- (B) the same substance (water) in three different states
- (C) three different chemicals that only look similar
- (D) only found on other planets

50. Steam rising from a hot pan of water cools near the ceiling and forms tiny droplets.

This change is, in order:

- (A) evaporation then condensation
- (B) condensation then evaporation
- (C) freezing then melting
- (D) precipitation then infiltration

51. Why is **only a small fraction** of Earth's huge water store available to us as drinking water?

- (A) because most of it is salty ocean water or frozen in ice
- (B) because the Sun dries it all up each day
- (C) because plants drink most of it
- (D) because it all flows to one country

52. During the dry season, a river shrinks and the village wells go deeper to find water.

This shows that in the dry season the water table has:

- (A) risen
- (B) fallen
- (C) turned to ice
- (D) become salty

53. Which of these does **not** raise the water table?

- (A) good monsoon rain on open soil
- (B) rooftop rainwater harvesting into recharge pits
- (C) over-pumping ground water for a new factory
- (D) building johads (small earthen check-dams) to hold rain

54. A region has scanty rainfall this year **and** a growing number of bore wells pumping ground water. The water table will most likely:

- (A) rise sharply
- (B) fall, because little water comes in while a lot is pumped out
- (C) stay exactly the same
- (D) turn into vapour

55. The chief reason farmers are encouraged to move from flood irrigation to drip irrigation is to:

- (A) make crops grow taller overnight
- (B) save water by cutting the large losses of flood irrigation
- (C) cool the soil
- (D) increase evaporation from the field

56. Which of the following is the correct chain for recharging ground water naturally?

- (A) rain falls → infiltrates open soil → reaches the aquifer → water table rises
- (B) rain falls → runs off concrete → into drains → water table rises
- (C) ground water is pumped out → soil dries → water table rises
- (D) sea water evaporates → falls as snow → water table rises

57. A colony replaces its lawns and soil paths with full concrete and tiles, then complains its wells dry up. The most likely reason is:

- (A) the concrete makes the water salty
- (B) the Sun got hotter
- (C) rain now runs off instead of soaking in, so the ground water is no longer recharged
- (D) the sea level rose

58. Compared with spraying water over a whole field, drip irrigation also reduces weeds because:

- (A) it floods the weeds away
- (B) only the crop's root zone is kept moist, so weeds between rows get little water
- (C) it sprays weedkiller
- (D) it freezes the soil

59. Which statement best captures the chapter's main message, "Water is a precious resource"?

- (A) water is unlimited, so we need not save it
- (B) usable fresh water is scarce, so we must use and manage it carefully
- (C) only farmers need to save water
- (D) the sea will always supply our drinking water

60. A town adopts all of these: rooftop rainwater harvesting, drip irrigation on nearby farms, fixing leaking pipes, and planting trees. Over a few years, the town's water table is most likely to:

- (A) fall faster than before
- (B) rise, because recharge improves and wasteful use drops
- (C) turn salty
- (D) stay exactly the same forever

Solutions — Science (Class 7), Chapter 16: Water: A Precious Resource

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	A	31	D	41	A	51	A
2	B	12	C	22	C	32	A	42	D	52	B
3	C	13	B	23	D	33	D	43	C	53	C
4	D	14	C	24	A	34	C	44	C	54	B
5	D	15	C	25	A	35	D	45	B	55	B
6	C	16	A	26	C	36	B	46	A	56	A
7	D	17	B	27	A	37	A	47	B	57	C
8	A	18	A	28	B	38	D	48	D	58	B
9	D	19	D	29	D	39	C	49	B	59	B
10	A	20	D	30	C	40	D	50	A	60	B

Answer-key distribution: A = 15, B = 15, C = 14, D = 16.

Worked Solutions

1. (B) Water exists in three **states**: solid (ice), liquid (liquid water) and gas (water vapour). The other lists mix in things that are not separate states — rain and snow are just falling liquid/solid water, "steam" and "clouds" are forms of vapour/droplets, and dew and fog are condensed liquid water. Only (B) names the three true states.

2. (B) When wet clothes dry in the Sun, the liquid water gains heat and changes straight into **water vapour** (evaporation). It does not freeze to ice, and it does not jump to clouds or rain — those are later stages of the cycle.

3. (C) The Sun's heat is what lifts water from oceans and land into the air, so **evaporation** is the Sun-powered step. Condensation and precipitation happen as the vapour cools and falls; infiltration is gravity pulling water into the soil — none of these needs the Sun directly.

4. (D) Dew is liquid water that appears when water vapour in the cool morning air touches cold grass and changes from gas to liquid — that is **condensation**. (It is the reverse of evaporation, and it is not falling precipitation.)

5. (D) Clouds are made of tiny water droplets. They form when **water vapour high in the cool air condenses** into those droplets. Rain falling, water freezing on the ground, or rivers reaching the sea are different events.

6. (C) Water vapour given off through the leaves of plants is **transpiration** — the plant's contribution to the water cycle. Evaporation is from open water/soil; condensation and precipitation are later stages.

7. (D) Rain, snow and hail are all forms of water falling from the sky, grouped together as **precipitation**. Evaporation and transpiration add vapour to the air; condensation makes clouds; only precipitation brings water back down.

8. (A) Water in the cycle keeps moving and changing state, but it is neither created nor destroyed — it only **changes its form and location**, so the total amount on Earth stays roughly constant. Rain does not "make" new water, evaporation does not destroy it, and it does not escape to space in any meaningful amount.

9. (D) The natural sequence is: the Sun evaporates water (evaporation) → the vapour cools and forms clouds (condensation) → the clouds release rain/snow (precipitation). So the order is **evaporation** → **condensation** → **precipitation**. The other orders put condensation or precipitation in the wrong place.

10. (A) The seas and oceans hold the overwhelming majority of Earth's water — **about 97%**. That is exactly why so little is left as usable fresh water. 50%, 2% or 0.5% badly underestimate the ocean share.

11. (D) Sea water cannot be drunk or used on crops mainly because it is **salty** — the dissolved salt harms us and most plants. Temperature, freezing or depth are not the core reason.

12. (C) The student's mistake is confusing "lots of water" with "lots of usable water." The correct fix: **most of Earth's water is salty or frozen, so the usable fresh water is very limited**. Earth does have plenty of water (so (B) is wrong), and sea water is not drinkable (so (D) is wrong).

13. (B) The largest part of Earth's *fresh* water is locked away — **frozen in ice-caps and glaciers, or stored deep underground** — so it is not easy to use. It is not all in the air, inside plants, or "inside ocean salt."

14. (C) The accurate statement is that **only a tiny fraction of Earth's water is fresh water we can readily use**. Most is salty ocean water; there certainly is salt water and glacier ice; so (A), (B) and (D) are all false.

15. (C) Having about 17% of the world's people but only about 4% of its fresh water means a large population must share a small water supply — India is **under water stress**. It is

not "more than enough" (A), it does have some fresh water (B), and exporting it (D) would worsen the shortage.

16. (A) Water that fills the spaces (pores) between soil particles and rocks beneath the surface is **ground water**. Surface water sits on top (rivers, ponds); sea water and rain water are different stores.

17. (B) The water table is the **upper level of the water stored underground** — the depth below which the ground is saturated. It is not the ocean top, a river, or a cloud layer.

18. (A) Good rain on open soil lets water infiltrate and recharge the ground water, so the saturated zone fills upward and the water table **rises**. It does not fall, vanish, or turn salty from rainfall.

19. (D) Rain water seeping down through the soil to join the ground water is **infiltration**. Evaporation and transpiration move water up as vapour; condensation makes clouds — none of those is downward seeping.

20. (D) An aquifer is **a layer of porous rock or gravel underground that stores water** in its pore spaces. It is not a surface lake, a pump, or a cloud.

21. (A) Ground water is recharged when **rain water infiltrates through the soil into the aquifer**. Releasing factory waste pollutes, pumping removes water, and ocean evaporation feeds clouds — none of these refills the aquifer.

22. (C) If the water pumped out each year exceeds what seeps back in, the underground store shrinks year on year, so the water table will **keep falling**. It cannot stay the same (more out than in), keep rising, or rebound on its own.

23. (D) The clearest long-term sign of a falling water table is that **hand-pumps and wells that once gave water now run dry** — the saturated level has dropped below them. A pond filling after rain, clouds forming, or dew on grass are short-term surface events, not signs of a falling table.

24. (A) **Pumping out large amounts of ground water for irrigation** removes water faster than it is replaced, lowering the table. Recharge pits, heavy rain on open ground and planting trees all **raise** or protect the table.

25. (A) Concreting over open ground stops rain from soaking in, so **less water recharges the ground water and the water table tends to drop**. Paving does not increase infiltration, raise the table, or salt the rain.

26. (C) All of growing population, paving with concrete, and intensive irrigation **lower** the water table. **Heavy rainfall soaking into open soil** does the opposite — it recharges and raises the table — so it is **not** a cause of a falling water table.

27. (A) Trees and their roots, plus the leaf litter beneath forests, slow run-off and help rain soak in. With fewer trees, **less rain water is held back and allowed to infiltrate, so recharge drops** and the table falls. The Sun, salt and rivers stopping are not the mechanism.

28. (B) A water table drops fastest when little comes in and a lot goes out. **Rising population, more industry, and heavy paving combined with scanty rainfall** maximise demand and minimise recharge. The other options are all conservation measures that protect or raise the table.

29. (D) Open soil lets rain infiltrate and recharge the ground water, while full concrete makes it run off. So after the monsoon the table is higher under **Colony X, because its open soil lets rain soak in**. Concrete does not trap or store ground water, and gardens do not "use up" the recharge advantage.

30. (C) The action that best *raises* a falling city water table is **building rainwater-harvesting recharge pits so rain soaks into the ground**. Paving parks, pumping more, and cutting trees all push the table down.

31. (D) Switching from flood to drip irrigation means the field now **uses much less water, easing the demand on ground water**. Drip does not waste more water, the crop still needs some water (not "none"), and it does not make ground water salty.

32. (A) Flood (surface) irrigation covers the whole field with water, so most of it evaporates or drains away unused — it wastes the most. Sprinklers waste less, and drip and hand-watering are the most economical.

33. (D) Drip irrigation saves water because it **delivers water drop by drop directly to each plant's root zone**, so almost none is lost to the bare soil between plants. Spraying high, flooding the basin, or watering only at night do not capture this targeting.

34. (C) To use the least water on a row of plants, a **drip system delivering water at each plant's roots** is best. Flooding the bed, an all-day sprinkler, or hosing the leaves all lose far more water.

35. (D) Rainwater harvesting means **collecting and storing rain where it falls, or letting it soak in to recharge ground water**. Buying bottled water, pumping faster, or draining rain to the sea are the opposite of harvesting.

36. (B) Leading rooftop rain through a pipe into an underground tank is **rooftop rainwater harvesting** — capturing the rain that falls on the building. It is not irrigation or deforestation.

37. (A) A recharge pit works by **letting collected rain water seep down into the aquifer**, topping up the ground water. It does not pump water out, seal the soil, or

evaporate water.

38. (D) A bawri (step well) and a tanka are traditional Indian water structures for **storing or harvesting rain and ground water** for later use. They do not generate electricity, cook, or measure rainfall.

39. (C) When rooftop rainwater harvesting becomes common, more rain is sent into the ground instead of running off, so over a few years the water table tends to **rise**. It does not fall sharply, turn salty, or freeze.

40. (D) World Water Day is observed on **22 March** each year. (5 June is World Environment Day; the others are unrelated.)

41. (A) Plants that do not get enough water lose firmness in their cells and **wilt — their leaves and stems droop**. Lack of water does not make them grow taller, vaporise, or flower more.

42. (D) Water scarcity hits people most directly by **making it hard to get enough water for drinking, cooking, washing and crops**. It does not give more water, cool the town, or raise the water table.

43. (C) Since even a slow drip wastes large amounts over a day, the right response is to **mend the leak promptly to stop the waste**. Ignoring it, opening the tap fully, or catching the drips forever all fail to save the water.

44. (C) **Using a bucket and mug instead of a long shower** uses the least water for a bath. A long shower, a running tap while soaping, and a full bathtub all use much more.

45. (B) Closing the tap while brushing **stops clean water from running away unused** down the drain. It has nothing to do with toothpaste working better, sea level, or the speed of the water cycle.

46. (A) **Rainwater harvesting, drip irrigation and mending leaks** all save fresh water. The other sets (paving, flood irrigation, leaving taps running; cutting forests, draining ponds, long showers; over-pumping, concreting, ignoring leaks) all waste it.

47. (B) To cut its demand on ground water, the school should **harvest rooftop rain into a recharge pit and fix all leaking taps** — adding supply and stopping waste. Drilling more bore wells, paving the playground, or running sprinklers overnight all increase use or reduce recharge.

48. (D) Comparing the stores, **ocean (salty) water is far more than the fresh water we can readily use** — roughly 97% versus a tiny percentage. Fresh usable water is **not** more than ocean water, not all water is fresh, and glacier ice is frozen, not our daily drinking source.

49. (B) Ice, liquid water and water vapour are **the same substance (water) in three different states** — solid, liquid and gas. They are not different substances or different chemicals.

50. (A) First the water in the hot pan turns to steam/vapour (**evaporation**); then that vapour cools near the ceiling and turns back into tiny droplets (**condensation**). So the order is evaporation then condensation — not the reverse, and not freezing/melting.

51. (A) Only a small fraction of Earth's water is drinkable **because most of it is salty ocean water or frozen in ice**. The Sun does not dry it all up, plants do not drink most of it, and it does not all flow to one country.

52. (B) In the dry season little water recharges the ground while use continues, so the water table has **fallen** — which is why the river shrinks and wells must be dug deeper. It has not risen, frozen, or turned salty.

53. (C) Good monsoon rain on open soil, rooftop rainwater harvesting, and check-dams (johads) all **raise** the water table. **Over-pumping ground water for a new factory** removes water, so it does **not** raise the table.

54. (B) With scanty rainfall (little recharge in) plus many bore wells (a lot pumped out), the water table will most likely **fall**. It cannot rise sharply, stay exactly the same, or turn into vapour under these conditions.

55. (B) Farmers are encouraged to move from flood to drip irrigation mainly to **save water by cutting the large losses of flood irrigation**. It does not grow crops overnight, cool the soil, or increase evaporation.

56. (A) The natural recharge chain is **rain falls → infiltrates open soil → reaches the aquifer → water table rises**. Run-off over concrete into drains does not recharge; pumping water out lowers the table; and sea-evaporation-to-snow is not the local recharge route.

57. (C) Replacing soil and lawns with concrete means **rain now runs off instead of soaking in, so the ground water is no longer recharged** and the wells dry up. The concrete does not salt the water, the Sun did not change, and the sea level is irrelevant here.

58. (B) Drip irrigation keeps **only the crop's root zone moist, so weeds growing between the rows get little water** and struggle. It does not flood weeds away, spray weedkiller, or freeze the soil.

59. (B) The chapter's main message is that **usable fresh water is scarce, so we must use and manage it carefully**. Water is not unlimited, saving it is everyone's job (not only farmers), and the salty sea will not supply our drinking water.

60. (B) Rooftop rainwater harvesting and recharge improve the inflow, while drip irrigation, fixing leaks and trees cut wasteful use and run-off. Together, over a few years, the town's water table is most likely to **rise**. It will not fall faster, turn salty, or stay frozen in place.

End of solutions. 60 questions, single correct option each. Key balanced A=15, B=15, C=14, D=16.